

Graduate Seminar Applied Logic (S4A6) WiSe 2026–27

Model Theory of Valued Fields

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Time and Place. Friday 10–12, SemR 0.007

Abstract. This seminar gives an introduction to the model theory of valued fields. Valued fields play a central role in modern model theory. The first-order behaviour of a valued field is often controlled by two simpler pieces of data: its residue field and its value group. This idea is made precise in Ax–Kochen–Ershov type principles and in quantifier elimination theorems such as those of Robinson, Macintyre, and Pas. The seminar gives an introduction to these themes. We begin with the basic algebraic theory of valued fields and ordered abelian groups, then discuss algebraically closed valued fields as well as the famous AKE principle. We then turn to the model theory of $\mathbb{Q}_p$, definability questions over p -adic fields, and transfer results for NIP. Depending on the interests of the participants, the final part of the seminar may also cover metric valued fields in continuous logic.

Rules.

- (1) Approximately one week before your talk, we will meet to discuss the content and structure of the presentation. Please prepare a first draft of your talk by then. Feel free to contact me earlier if mathematical or organisational questions arise.
- (2) In addition to giving a 90-minute talk, each participant who wants to receive credit is required to submit a written summary of at least four pages on their topic by the time of their talk.
- (3) Your grade will be determined, among other things, by the mathematical content of your talk, your understanding of the material, the pedagogical effort, the quality of the presentation, and the written summary. Your preparation for the meeting before the talk will also be taken into account.

Prerequisites. This seminar is designed for students who have already taken a first course in model theory, for example Mathematical Logic (V3A5/F4A1–5) during the Winter Semester 2025/26. In addition, only knowledge from a basic algebra course (covering e.g. field extension, ideals,...) is assumed.

Participants are however highly encouraged to make themselves acquainted with basic notions around valued fields (e.g. Section 1–3.1 of [7]) beforehand which will be reviewed during the first two talks.

Talks.**(1) Overview and introduction to valued fields I** (16.10.26 - *Stefan Ludwig*)

These introductory talks give an overview of the basic algebraic theory of valued fields, following Sections 1–3.1 of [7]. They should cover valuations, valuation rings, residue fields, value groups, examples, extensions of valued fields, and the basic structure of henselian fields. The talks are intended as a condensed summary of the background material. Participants are strongly encouraged to read [7] beforehand.

(2) Overview and introduction to valued fields II (23.10.26 - *Stefan Ludwig*)

Continuation of the first talk and 4.1–4.6 of [7].

(3) Hensel’s Lemma and Henselianisation (30.10.26 - *Lovre Pazinović*)

Cover 3.10 in [7] with proof (idea) as well as Exercise 3.14. Cover 4.8–4.13 prioritising the proof of 4.13.

(4) Ordered abelian groups (06.11.26 - *Elena Buchholz*)

Recall the model theory of divisible ordered abelian groups and Presburger arithmetic, including the corresponding quantifier elimination results; see, for example, Section 3 of [6]. Briefly explain the relationship between convex subgroups of the value group and coarsenings of valuations; see Section 2.3 of [10] or [4]. Finally, introduce NIP and state the theorem of Gurevich and Schmitt that ordered abelian groups are NIP [5]. If time permits, give a sketch of the proof.

(5) Algebraically closed valued fields I: quantifier elimination (13.11.26 - *Arthur Sliwa*)

Introduce algebraically closed valued fields and present Robinson’s quantifier elimination theorem for ACVF, following Section 4 of [6]. The statement and proof of Theorem 4.4 should be the central part of the talk.

(6) Algebraically closed valued fields II: definable sets and NIP (20.11.26 - *Alexandros Ritsogiannis*)

Continue the study of ACVF by discussing the description of definable sets, especially Corollary 4.11 of [6]. If time permits, also explain why ACVF is NIP, following Theorem 4.14 of [6].

(7) Kaplansky theory (27.11.26 - *Joa Fiege*)

The goal of this talk is to present the algebraic ingredients needed for Pas’ quantifier elimination in the following talk. The choice of results should be guided by this goal. In particular, the talk should cover Theorem 5.14 and Theorem 5.24 of [10]. Alternatively, the speaker may follow Section 5 of [6].

(8) The Ax–Kochen–Ershov theorem and Pas’ quantifier elimination (04.12.26 - *Jiayang Zhang*)

Present the beginning of Section 6 of [6], up to and including Pas’ quantifier elimination theorem.

(9) Applications of Ax–Kochen–Ershov (11.12.26 - *Lars Nießen*)

Start with Corollary 6.11 of [6]. Then present Theorem 6.17 of [6]. Finally, give an application to Artin’s conjecture using the Ax–Kochen–Ershov theorem, following Section 6.3 of [10].

(10) **NIP transfer for valued fields** (18.12.26 - Robin Selber)

The main result to present is Delon’s theorem: a henselian valued field of residue characteristic zero is NIP if and only if its residue field is NIP. See Appendix A.15 and the subsequent proof in [12]. If time permits, the speaker may also briefly discuss the main result of [8].

(11) **Some model theory of $\mathbb{Q}_p$** (15.01.27 - Sergio Esperalta Delgado)

Present quantifier elimination for $\mathbb{Q}_p$ in Macintyre’s language, including the proof. One possible source is Section 7.1 of [10]; the original paper is [9].

(12) **Metric valued fields in continuous logic** (22.01.27 - Nguyen Nam Hung Tran)

Give a brief introduction to the syntax of continuous logic, following Section 3 of [2]. Then present Ben Yaacov’s formalization of metric valued fields in continuous logic, covering [1, Section 1.1–Theorem 1.8].

References

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